Unit Choice Supplement for Studies in Probability and Statistics

We outline three pathways for unit choices involving probability and statistics. This document serves as a supplement to the document “Guide to making Your Unit Choices for 2014-15”. The purpose of this document is to give perspective to students on how their unit choices may work together throughout their studies. Please always consult the “Prerequisite Flowcharts” and other available information as well, this document is not intended as a replacement to any of those. The three respective pathways are designed to provide coherent and strong foundation in

- Probability Theory and Stochastic Processes;
- Decision Making, Computational Statistics and Optimization;
- Statistical Modelling and Inference.

Units typed in **red** are essential and indispensable. In particular, the earlier ones (like *Metric Spaces* and *Measure Theory and Integration* in the Probability Theory and Stochastic Processes pathway) contain fundamental prerequisites without which later, more advanced units cannot be efficiently followed.

Units typed in **green** are still very important for a mathematician with specialization in the respective pathway. We suggest these ones as further optional – but still strongly recommended – choices. Some units which are available both on levels H/6 and M/7 are featured twice in our list. A few of these are green in H/6 and red in M/7 meaning that we think it essential and indispensable by Year 4, but can already be taken at an earlier level.

Probability Theory and Stochastic Processes

This pathway is intended for students interested in a more detailed understanding of probability theory and stochastic processes. It also lists supplementary courses (mostly from analysis) which we think are essential for a solid foundation. Completing this pathway will make anyone familiar with the basic concepts and techniques that are in the heart of the theory of probability and stochastic processes. Today this area forms an essential part of pure mathematics, and therefore this pathway is a valuable step towards frontier-line research in pure mathematics as well as applied probability.

LEVEL I/5, 1ST TEACHING BLOCK

Metric Spaces (20cp)
Statistics 2 (20cp)
Multivariable Calculus (10cp)

LEVEL I/5, 2ND TEACHING BLOCK

Applied Probability 2 (20cp)
Ordinary Differential Equations (20cp)
Combinatorics (20cp)

LEVEL H/6, 1ST TEACHING BLOCK

Measure Theory and Integration (20cp)
Complex Networks (20cp)
Complex Function Theory (20cp)
Dynamical Systems and Ergodic Theory (20cp)
Introduction to Queuing Networks (10cp)
Random Matrix Theory (10cp)

LEVEL H/6, 2ND TEACHING BLOCK

Further Topics in Probability (20cp)
Martingale Theory with Applications (10cp)
Functional Analysis (20cp)
Financial Mathematics (20cp)

LEVEL M/7, 1ST TEACHING BLOCK

Introduction to Stochastic Analysis (20cp)
Dynamical Systems and Ergodic Theory (20cp)
Complex Function Theory (20cp)
Introduction to Queuing Networks (10cp)

LEVEL M/7, 2ND TEACHING BLOCK

Financial Mathematics (20cp)
Functional Analysis (20cp)

Decision Making, Computational Statistics and Optimization

This collection of units is suitable for students interested in topics which straddle the boundary between statistics and applied probability. The topics covered here will provide students with insight into a variety of modelling techniques, associated computational methods and the mathematical tools needed for their rigorous understanding and analysis. The key starting points, highlighted in **red** are the Statistics 2 and Applied Probability 2 units.

LEVEL I/5, 1ST TEACHING BLOCK

- **Statistics 2 (20cp)**
- **Methods of Complex Functions (10cp)**
- **Multivariable Calculus (10cp)**
- **Metric Spaces (20cp)**
- **Optimization (20cp)**

LEVEL I/5, 2ND TEACHING BLOCK

- **Applied Probability 2 (20cp)**
- **Ordinary Differential Equations (20cp)**

LEVEL H/6, 1ST TEACHING BLOCK

- **Bayesian Modelling A (10cp)**
- **Introduction to Queuing Networks (10cp)**
- **Linear Models (10cp)**
- **Measure Theory and Integration (20cp)**

LEVEL H/6, 2ND TEACHING BLOCK

- **Bayesian Modelling B (10cp)**
- **Financial Mathematics (20cp)**
- **Information Theory (10cp)**
- **Martingale Theory with Applications (10cp)**
- **Theory of Inference (20cp)**

LEVEL M/7, 1ST TEACHING BLOCK

- **Complex Networks (20cp)**
- **Generalized Linear Models (10cp)**
- **Monte Carlo Methods (10cp)**
- **Stochastic Processes (10cp)**

LEVEL M/7, 2ND TEACHING BLOCK

- **Evolutionary Game Theory (10cp)**
- **Functional Analysis (20cp)**
- **Graphical Models (10cp)**
- **Stochastic Optimization (10cp)**

Statistical Modelling and Inference

This collection of units is suitable for students interested in statistical approaches to modelling, analysis and interpretation of data, and the mathematical foundations of these techniques. The key starting point is the Statistics 2 unit; beyond that there are a variety of possible combinations of units, some more practical, some more theoretical.

LEVEL I/5, 1ST TEACHING BLOCK

- **Statistics 2 (20cp)**
- **Methods of Complex Functions (10cp)**
- **Multivariable Calculus (10cp)**
- **Optimisation (20cp)**

LEVEL I/5, 2ND TEACHING BLOCK

- **Applied Probability 2 (20cp)**
- **Applied Partial Differential Equations (20cp)**

LEVEL H/6, 1ST TEACHING BLOCK

- **Bayesian Modelling A (10cp)**
- **Linear Models (10cp)**
- **Multivariate Analysis (10cp)**
- **Time Series Analysis (20cp)**

LEVEL H/6, 2ND TEACHING BLOCK

- **Bayesian Modelling B (10cp)**
- **Experimental Design (10cp)**
- **Theory of inference (20cp)**

LEVEL M/7, 1ST TEACHING BLOCK

- **Generalized Linear Models (10cp)**
- **Monte Carlo methods (10cp)**
- **Nonparametric regression (10cp)**

LEVEL M/7, 2ND TEACHING BLOCK

- **Advanced time series (10cp)**
- **Financial Mathematics (20cp)**
- **Graphical Models (10cp)**